

Home Work 3

II B. Tech. (Common)

Mathematics-II (NMCI102)

Notations:

- $M_{m \times n}(\mathbb{F})$: The set of all matrices of order $m \times n$ with entries from $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.
 $M_n(\mathbb{F})$: The set of all matrices of order $n \times n$ with entries from $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.
 $P(\mathbb{R})$ or $\mathbb{R}[x]$: The set of all polynomials in one variable with real coefficients.
 $P_n(\mathbb{R})$ or $\mathbb{R}_n[x]$: The set of all polynomials of degree at most n in one variable with real coefficients.
 $\mathbb{R}^\infty$: The set of all real-valued sequences.
 $\mathcal{F}(X, \mathbb{R})$: The set of all real-valued functions on $X \subseteq \mathbb{R}$.
 $C(\mathbb{R})$: The set of all real-valued continuous functions on $\mathbb{R}$.
 $C[0, 1]$: The set of all real-valued continuous functions defined on $[0, 1]$.
 $D(0, 1)$: The set of all real-valued differentiable functions defined on $(0, 1)$.

- Which of the following subsets are subspaces over $\mathbb{R}$ with respect to the usual addition and scalar multiplication.

(i) $\{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) \mid f(x^2) = f(x)^2\}$	(ii) $\{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) \mid f(1) = 0\}$
(iii) $\{A \in M_n(\mathbb{C}) \mid A \text{ is non-invertible}\}$	(iv) $\{A \in M_3(\mathbb{R}) \mid A \text{ is a nilpotent matrix}\}$
(v) $\{(x, y) \in \mathbb{R}^2 \mid (x - y)(x + y) = 0\}$	(vi) $\{(x, y, z) \in \mathbb{R}^3 \mid x = z \}$
(vii) $\{(x, y, z) \in \mathbb{R}^3 \mid x = 0 \text{ or } y = 0\}$	(viii) $\{p(x) \in P_n(\mathbb{R}) \mid p(x) = p(1 - x)\}$
(ix) $\{A \in M_n(\mathbb{C}) \mid \bar{A}^T = A\}$	(x) $\{A \in M_n(\mathbb{C}) \mid \bar{A}^T = -A\}$
(xi) $\{(x, y) \in \mathbb{R}^2 \mid y \leq x\}$	(xii) $\{(x, y) \in \mathbb{R}^2 \mid y = x \}$
(xiii) $\{(x, y) \in \mathbb{R}^2 \mid y \leq x\}$	(xiv) $\{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) \mid f(x) = a \cos x + b \sin x + c, a, b, c \in \mathbb{R}\}$
- Which of the following are subspaces of the vector space $\mathbb{R}^n$?

(a) $S = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : x_n = 0\}$ (b) $S = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : x_1 + x_2 + \dots + x_n = 0\}$
 (c) $S = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : x_1^2 + x_2^2 + \dots + x_n^2 \geq 1\}$ (d) $S = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : x_i = x_{n-i+1} \forall i = 1, 2, \dots, n\}$.
- Which of the following are subspace of $\mathbb{R}[x]$?

(a) $S = \mathbb{R}_n[x]$ (b) $S = \{f(x) \in \mathbb{R}[x] : f(x) = f(1 - x) \forall x\}$ (c) $S = \{f(x) \in \mathbb{R}[x] : f(x) = f(-x) \forall x\}$
 (d) $S = \{f(x) \in \mathbb{R}[x] : f(1) \geq 0\}$ (e) $S = \{f(x) \in \mathbb{R}[x] : f'(0) + f(0) = 0\}$ (f) $S = \{f(x) \in \mathbb{R}[x] : f(x) \text{ has a root in } [-1, 1]\}$.
- Which of the following are subspaces of the vector space $M_{2 \times 2}(\mathbb{R})$?

(a) $S = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : a + b = 0 \right\}$ (b) $S = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : a + b + c + d = 0 \right\}$
 (c) $S = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = 0 \right\}$ (d) $S = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : b = c = 0 \right\}$
 (e) $S = \{A \in M_{2 \times 2}(\mathbb{R}) : A = A^T\}$ (f) $S = \{A \in M_{2 \times 2}(\mathbb{R}) : A = -A^T\}$
 (g) $S = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : c = 0 \right\}$ (h) $S = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : b = 0 \right\}$.

Answers: (a) Yes (b) Yes (c) No (d) Yes (e) Yes (f) Yes (g) Yes (h) Yes.

- Which of the following are subspaces of the vector space $\mathcal{F}([0, 1], \mathbb{R})$?

(a) $S = C[0, 1]$ (b) $S = \{f \in C[0, 1] : f(0) = 0\}$ (c) $S = \{f \in C[0, 1] : f(0) = 0, f(1) = 0\}$ (d) $S = D(0, 1)$
 (e) $S = \{y \in D(0, 1) : \frac{dy}{dx} + p(x)y = 0\}$ (f) $S = \{y \in D(0, 1) : \frac{dy}{dx} + p(x)y = q(x), q(x) \neq 0\}$.

Answers: (a) Yes (b) Yes (c) Yes (d) Yes (e) Yes (f) No..

- For $A = \begin{bmatrix} 1 & 2 & 5 \\ 3 & 0 & 7 \\ -1 & 4 & 3 \end{bmatrix}$, examine whether $(1, 1, 1)$ and $(1, -1, 1)$ are in (a) the row space of A ; (b) the column space of A .

Answers: (a) no, yes (b) yes, no

7. Determine whether $(1, 1, 1) \in L\{(1, 3, 4), (4, 0, 1), (3, 1, 2)\}$.
8. Express the polynomial $p(t) = t^2 + 4t - 3$ over $\mathbb{R}$ as a linear combination of the polynomials $p_1(t) = t^2 - 2t + 5$, $p_2(t) = 2t^2 - 3t$ and $p_3(t) = t + 3$, if possible.
9. Write TRUE/FALSE with proper justifications.
- (a) Any set containing the zero vector is linearly dependent
 - (b) If S is a linearly dependent set, then each vector in S is a linear combination of other vectors in S
 - (c) Subsets of linearly independent sets are linearly independent
 - (d) Subsets of linearly dependent sets are linearly dependent.
- Answers:** (a) TRUE (b) FALSE (c) TRUE. (d) FALSE
10. Determine the linear independence/dependence of the following sets in the corresponding vector spaces.
- (a) $\{x^3 + 2x^2, -x^2 + 3x + 1, x^3 - x^2 + 2x - 1\}$ in $P_3(\mathbb{R})$
 - (b) $\{(1, 2, 2), (2, 1, 2), (2, 2, 1)\}$ in $\mathbb{R}^3$
 - (c) $\left\{ \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix}, \begin{bmatrix} -2 & 6 \\ 4 & -8 \end{bmatrix} \right\}$ in $M_{2 \times 2}(\mathbb{R})$
 - (d) $\{(1, 2, 1, -2), (2, 5, -1, 3), (1, 3, -2, 5), (3, 1, 2, -4)\}$ in $\mathbb{R}^4$.
- Answers:** (a) LI (b) LI (c) LD (d) LD.
11. For what real values of k , the set $\{(k, 0, 1), (1, k + 1, 1), (1, 1, 1)\}$ is a LI subset of $\mathbb{R}^3$? [**Ans.** $k \neq 0, 1$]
12. Let u and v be distinct vectors in any vector space V over F . Show that $\{u, v\}$ is linearly dependent if and only if u or v is a multiple of the other.
13. Let $\{u, v, w\}$ be linearly independent in a real vector space V . Show that $\{\lambda u, \lambda v, \lambda w\}$, $\lambda \neq 0$, $\{u + \lambda v, v, w\}$, $\{u + v, u + w, v + w\}$, $\{u + v + w, v + w, w\}$ are also linearly independent in V and $\{u + \lambda v, v + \lambda w, w + \lambda u\}$ may not be linearly independent in V , where $\lambda \in \mathbb{R}$.
14. Let S be a subset of a vector space V over a scalar $F = \mathbb{R}$ or $\mathbb{C}$. Show that $L(S)$ is the smallest subspace of V containing S .